

fat (SNF), fatty acids (FAT) details using the farmer's RFID tags. The details are sent to the farmer via SMSes. Data acquired is used by Cloud-side applications to determine return on investment improvement, enhanced traceability, regional assessment of milk production pattern and automated real-time farmer payments.

At storage units. Milk cold-chain-side IoT intervention includes sensors for accessing milk temperature, volume of milk, energy optimisation and pilferage control. Data acquired is used by Cloud-side application to ensure adherence to the cold chain protocol, determine quality of milk and for preventive maintenance of milk-chilling equipment.

Putting agriculture and technology together

Agriculture, by default, is rural and remote. This causes troubles like remote management and difficulty in getting local expertise on premise. This is a major problem in emerging markets especially where farm productivity levels, quality of farm produce and production-side supply chain values are extremely low as compared to developed markets.

Stellapps team analysed this issue and found out that using sensors in the farms to automatically acquire data and apply analytics and machine learning on the Cloud and then using the outcome of the analytics and machine learning can solve both the issues efficiently.

A simple solution might be implanting a sensor on the milking equipment, storage units and pedometers, and deploying a milk-collection centre with milk-examining equipment. Connect all devices through the Cloud, collect data and send it to the dairy. However, this is not all.



Fig. 3: SmartMoo milk procurement application



Fig. 4: The founding team behind SmartMoo (left to right: Ramakrishna Adukuri, Venkatesh Seshasayee, Ranjith Mukundan, Ravishankar G. Shiroor and Praveen Nale)

Because the whole team were from IT and technology background, it took them a lot of time to understand agriculture and dairy verticals. "This lack of vertical experience was a disadvantage as well as an advantage. We were able to look at the problems from a fresh perspective," says the Stellapps team.

The major problem was to get customers to appreciate the need and value of the product, and obviously the major customers, farmers and the rural crowd found it difficult to digest too much technology.

Once the farmers understood and accepted the need, the next challenge arose—remoteness of deployments made it hard for support and maintenance. This challenge has been overcome today by providing most of the support via Cloud interfaces and also via training customer support teams.

Another issue is that customers expect software Cloud services to be free, which is a problem the team is trying to surmount.

The bigger picture

The whole company is bootstrapped using a corpus from career savings of the founders. The founding team

members have strong technology-industry experience and come from IITs. In addition to this, a good technology-centric idea made it easy to get funding from various alumnus including IIT-Madras/ Rural Technology Business Incubator (RTBI). "We were lucky to run into Omnivore Partners, who had agriculture-technology-specific focus, and Stellapps' business vision aligned with Omnivore's," says the team.

The whole system does not benefit the farmers and the company alone but also ensures end customers with good-quality products. "We are touching more than

350,000 farmers every day through our solutions and around 4.7 million litres of milk every single day," the team adds.

The proprietary smartSDP platform and suite of apps touch close to two billion litres of milk annually. Proprietary solutions are deployed successfully at many private dairies in India, Kenya and Nepal as they continue to rapidly expand. Not just this, the response of SmartMoo users is also overwhelming—a lot of farmers from various districts of Karnataka like Kolar, Coorg, Mysore and others find SmartMoo very useful.

Through Stellapps technology solutions applied at the Nepal farm, there has been a substantial increase in productivity of the farm, touching 800 cattle, and the farm has been able to draw upon several thousands of litres of milk with clean milk technology and raise the health and monitoring services of cattle.

SmartMoo's centre of gravity is shifted towards software/data on the Cloud, thereby commoditising the hardware. This Bengaluru based startup aims to create a major revolution in the agriculture industry by revamping the dairy farms to boost productivity. **EFY**